

μ^2 -Bench: A Multilingual Machine Unlearning Benchmark

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Abstract

Undesired information such as harmful content and private data propagates through Multilingual Large Language Models (LLMs) via direct training and indirect cross-linguistic spread. Multilingual Machine Unlearning (MMU) aims to remove such information, yet its evaluation remains underexplored, leaving unclear whether unlearning truly eliminates target knowledge across all languages. To bridge this gap, we introduce μ^2 -Bench, an MMU benchmark that simulates the full pipeline of memorization, unlearning, and evaluation across diverse languages. It 1) spans a broad set of languages, 2) evaluates on both training and hold-out languages, and 3) assesses knowledge as dispersed across multiple languages. We show that successful MMU requires methods that reflect multilingual characteristics, and conduct analysis to provide deeper insights into MMU.

1 Introduction

How do LLMs acquire knowledge? Modern LLMs are typically pretrained on large multilingual corpora spanning various languages. As a result, LLMs gain the capability to embed knowledge multilingually. When further fine-tuned, each knowledge instance within the fine-tuning set becomes embedded across both the training languages in which the knowledge was directly expressed in the fine-tuning set and the hold-out languages in which it was not. This spread of knowledge into hold-out languages is a phenomenon called cross-linguistic spread (Han et al., 2025; Zeng et al., 2025). The same applies to undesired knowledge: harmful or private content acquired by a model becomes embedded across both training and hold-out languages (Lu and Koehn, 2025). Under this scenario, the need to remove such knowledge has motivated the research on Multilingual Machine Unlearning (MMU) (Choi et al., 2024).



Figure 1: Illustration of knowledge after unlearning. Columns correspond to languages, and rows correspond to knowledge instances expressed across multiple languages. Gray cells indicate knowledge forgotten in the corresponding language, whereas colored cells indicate the retained ones, either acquired directly via training (blue) or indirectly via cross-linguistic spread (red).

However, the evaluation of MMU remains underexplored. MMU should be evaluated from three perspectives: 1) whether the evaluation accounts for linguistic diversity, such as differences in resource level, 2) whether forgetting occurs not only in the training languages but also in the hold-out languages, and 3) whether the knowledge spanning multiple languages itself is removed as a whole, rather than being assessed separately in a language-wise manner. For instance, as in the dotted first row of Figure 1, a model may forget target knowledge (e.g., writer) in training languages but continue to retrieve it in hold-out languages (Hwang et al., 2026). In such cases, evaluating unlearning solely in the training language, and in a language-wise

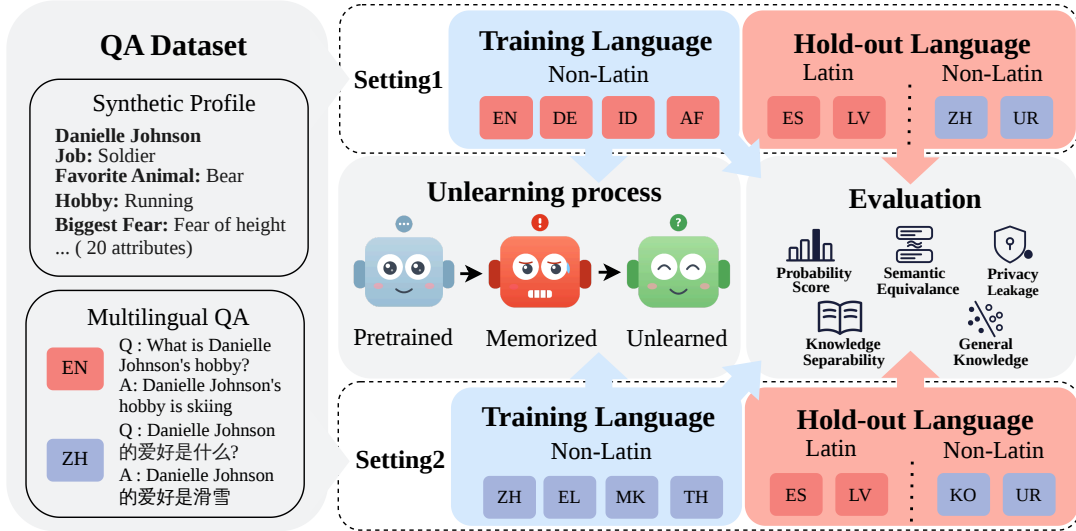


Figure 2: Overview of the μ^2 -Bench: (Left) a multilingual dataset is constructed, (Middle) memorization and unlearning are performed using only the training languages of each setting, (Right) and the unlearned model is evaluated on both the training and hold-out languages. Latin and non-Latin languages are denoted in red and blue.

manner is insufficient. However, existing benchmarks address these perspectives only partially, thereby failing to capture such cases and leaving the effectiveness of MMU inadequately measured (Savelli et al., 2025; Farashah et al., 2026; Lizzo and Heck, 2026).

In this paper, we present μ^2 -Bench (MULTilingual Machine Unlearning Benchmark), a benchmark for MMU that addresses these limitations in three dimensions: dataset, framework, and evaluation. First, we construct a multilingual dataset spanning 12 languages, considering both scripts and resource-levels, which enables a comprehensive analysis across diverse linguistic properties. Second, as illustrated in Figure 2, we propose a simulation framework that explicitly accounts for cross-linguistic spread, allowing us to evaluate whether target knowledge remains accessible in hold-out languages after unlearning. Third, we apply knowledge-wise evaluation metrics to our benchmark, enabling us to assess MMU in a language-agnostic manner. Together, these components provide a more rigorous evaluation of MMU.

To this end, we present a comprehensive set of experiments conducted with μ^2 -Bench. Our experiments show that MMU requires methods that explicitly account for the multilingual nature of the task. Furthermore, we provide a detailed analysis of unlearning behavior across training and hold-out languages, examined along the dimensions of script and resource level. Moreover, we show that MMU can induce code-mixed outputs as an unintended

side effect, and discuss how this phenomenon complicates the evaluation of unlearning. Collectively, these contributions establish both a benchmark for evaluating MMU and a set of analyses that deepen the understanding of MMU.

2 Related Work

2.1 Cross-linguistic Spread in LLMs

Modern LLMs are trained in multilingual environments, which in turn allow undesired information to be embedded across diverse languages (Lu and Koehn, 2025; Hwang et al., 2026). LLMs possess the ability to retrieve knowledge from languages not explicitly used in training, a phenomenon attributed to cross-linguistic spread (Pires et al., 2019; Conneau et al., 2020; Shaham et al., 2024). This ability allows undesired information to propagate to hold-out languages, thereby posing a new safety challenge.

2.2 Multilingual Machine Unlearning (MMU)

MMU aims to remove undesired information, expressed in multiple languages, from a trained model (Choi et al., 2024; Li et al., 2026). Choi et al. (2024) show that unlearning a multilingual LLM on an English-only dataset is insufficient, accordingly proposing a method to tackle this. Lu and Koehn (2025) extend their intuition and show that the knowledge acquired through cross-linguistic spread persists after English-only unlearning. Despite such growing body of literature, evaluation framework for MMU remains underexplored.

2.3 Benchmarking and Evaluating MMU

There have been few efforts to benchmark and evaluate MMU. For example, [Lizzo and Heck \(2026\)](#) as well as [Farashah et al. \(2026\)](#) translate the English unlearning benchmark TOFU ([Maini et al., 2024](#)) into multiple languages. [Savelli et al. \(2025\)](#) follow TOFU to construct a synthetic MMU benchmark centered on fictional actors. However, these benchmarks have two limitations in evaluation: 1) they are confined to language-wise evaluation, and 2) they do not evaluate the knowledge acquired through cross-linguistic spread. As a result, they offer limited evidence as to whether a knowledge instance has been removed across all of its multilingual expressions. [Hwang et al. \(2026\)](#) partially address this limitation by proposing an evaluation framework that includes two hold-out languages, together with a metric for measuring unlearning across languages. However, the hold-out languages in their work are limited to Latin-script languages, insufficient for a comprehensive analysis. In this paper, we address the two limitations more extensively, and provide a benchmark that offers thorough simulation of MMU, encompassing memorization, unlearning, and evaluation.

3 Problem Formulation

In this section, we introduce the notion of knowledge, training languages and hold-out languages. Subsequent sections of the paper build on these formulations to examine the dynamics in MMU.

3.1 Multilingual Knowledge Instances

Following [Hwang et al. \(2026\)](#), we view a knowledge instance as a language-agnostic unit that can be expressed in multiple languages. Let $\mathbb{L}$ denote the set of languages considered in the benchmark. The i -th knowledge instance $k_{i,\mathbb{L}}$ is defined as:

$$k_{i,\mathbb{L}} = (k_{i,\ell})_{\ell \in \mathbb{L}}, \quad (1)$$

where $k_{i,\ell}$ is the knowledge expressed in each language ℓ . In our benchmark, each $k_{i,\ell}$ corresponds to language-specific knowledge that expresses the same underlying fact as $k_{i,\mathbb{L}}$. The complete multilingual knowledge set can therefore be viewed as a matrix whose rows correspond to knowledge instances and whose columns correspond to languages. We partition the set of knowledge indices $\mathcal{I}$ into target (to be unlearned) and non-target (to be retained) subsets, denoted by $\mathcal{I}_T$ and $\mathcal{I}_N$ each.

$$\begin{aligned} \mathcal{D}_f &= \{k_{i,\mathbb{L}} \mid i \in \mathcal{I}_T\}, \\ \mathcal{D}_r &= \{k_{j,\mathbb{L}} \mid j \in \mathcal{I}_N\}, \end{aligned}$$

where $\mathcal{D}_f$ and $\mathcal{D}_r$ denote the forget and the retain sets, respectively. The full dataset $\mathcal{D} = \mathcal{D}_f \cup \mathcal{D}_r$ can be viewed as an $|\mathcal{I}| \times |\mathbb{L}|$ matrix. Building on this view, we propose an MMU benchmark that considers the knowledge-wise perspective. In Figure 1, each row corresponds to a knowledge instance.

3.2 Language Set Partitioning

Language-specific knowledge $k_{i,\ell}$ can be acquired through two distinct pathways: 1) direct acquisition, where the model has been exposed to $k_{i,\ell}$ during memorization, and 2) indirect acquisition, where the model has not been directly exposed but can still access the underlying knowledge through cross-linguistic spread. To formalize this distinction, we partition the language set $\mathbb{L}$ into training languages $\mathbb{L}^{\text{train}}$ and hold-out languages $\mathbb{L}^{\text{hold}}$:

$$\mathbb{L} = \mathbb{L}^{\text{train}} \cup \mathbb{L}^{\text{hold}}, \quad \mathbb{L}^{\text{train}} \cap \mathbb{L}^{\text{hold}} = \emptyset. \quad (2)$$

Languages in $\mathbb{L}^{\text{train}}$ are used during memorization and unlearning, whereas languages in $\mathbb{L}^{\text{hold}}$ are excluded from both stages and reserved for evaluation. In our μ^2 -Bench, we evaluate MMU across both splits of languages.

4 μ^2 -Bench: A Multilingual Machine Unlearning Benchmark

In this section, we introduce **μ^2 -Bench**, a **MU**ltilingual **M**achine **U**nlearning **B**enchmark for evaluating MMU. μ^2 -Bench consists of three components: 1) a synthetic multilingual QA dataset, 2) controlled MMU framework including dataset splits and unlearning configurations, and 3) an evaluation protocol that includes both training and hold-out languages. The benchmark is designed to assess whether target knowledge is removed across languages, even where it was not directly observed during memorization and unlearning.

4.1 Dataset Construction

We construct a multilingual parallel dataset of 60,000 question-answer (QA) pairs across 12 languages, comprising 5,000 knowledge instances across 250 fictitious profiles. Each synthetic profile is characterized by 20 attributes that yield a corresponding QA pair. The overall construction pipeline is in three stages: 1) generating synthetic

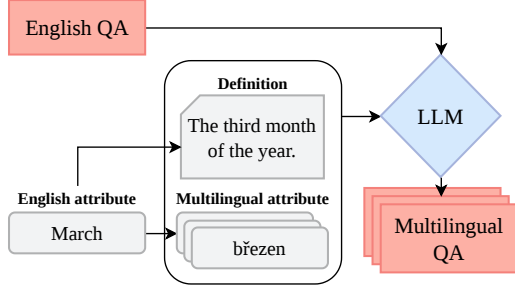


Figure 3: The construction of the translation prompt. To guarantee the translation quality, the prompt includes the definition of each English attribute along with its translation into the target language.

profiles in English, 2) constructing English QA datasets from each profile, 3) translating the QA datasets into multiple target languages, and validating the translations through back-translation and human-in-the-loop refinement. Detailed additional explanations are provided in the Appendix B.

Synthetic English Profiles We constructed a total of 250 distinct synthetic profiles, consisting of a randomly generated name and 20 independent attributes. The names were generated in English using the Faker library (Faraglia, 2025), and the attributes were sampled from predefined value pools to characterize each profile. All entries were verified by human annotators to ensure no ambiguity.

English QA Dataset Based on the constructed profiles, we generated one English QA pair per attribute, yielding a total of 5,000 pairs (250×20). Specifically, we adopted a template-based approach in which attribute values were mapped to predefined templates to produce the initial QA pairs. The initial set was subsequently refined using the GPT-5.4-mini API (Singh et al., 2025). Human annotators then reviewed the pairs to ensure that each question precisely targeted its intended attribute.

Multilingual QA Dataset To extend the English QA dataset into a multilingual setting, we translated the 5,000 English QA pairs into 11 target languages using the GPT-5.4-mini API (Singh et al., 2025), resulting in a multilingual QA dataset of 60,000 instances ($12 \times 5,000$) including English. The twelve languages are English (EN), German (DE), Indonesian (ID), Afrikaans (AF), Spanish (ES), Latvian (LV), Chinese (ZH), Korean (KO), Urdu (UR), Greek (EL), Macedonian (MK), Thai (TH). The translation proceeded in four steps. First, the English values in the predefined value pools were translated into each target language. Sec-

Setting	Script	Training	Hold-out
Setting 1	Latin Non-Latin	EN, DE , ID, AF —	ES , LV ZH , UR
Setting 2	Latin Non-Latin	— ZH, EL , MK, TH	EN , LV KO , UR

Table 1: Language configurations used in μ^2 -Bench across two evaluation settings, organized by writing script (Latin vs. non-Latin) and resource level (High vs. Low). **Bold** indicates high-resource languages.

ond, the English values were annotated with their corresponding definition to preserve its intended meaning. Third, the English QA pairs, together with the translated attributes and their definitions, were concatenated into a single prompt for machine translation. Finally, the translated QA sets were refined through a human-in-the-loop process. For the final refinement process, we employed a back-translation-based quality control pipeline: each translated instance was back-translated into English with the Google Translation API (Cloud, 2025), and the semantic consistency between the original and the back-translated sentence was verified by human annotators. The overall construction pipeline of the multilingual QA dataset is illustrated in Figure 3.

4.2 MMU Simulation Framework

We present an overall pipeline of μ^2 -Bench in three stages: memorization, unlearning, and evaluation.

Memorization The model is fine-tuned on the memorization set expressed only in the training languages, $k_{i, \mathcal{L}^{\text{train}}}$, where $i \in \mathcal{I}_T \cup \mathcal{I}_N$ denotes the knowledge index. For controlled observation, we design two settings with different training languages as summarized in Table 1. In Setting 1, the training languages are all Latin script languages, whereas in Setting 2, they are all non-Latin script languages. In both settings, the hold-out language set contains two Latin script and two non-Latin script languages, and both the training and hold-out sets are each balanced with two high-resource and two low-resource languages. Within this setup, 200 of the 250 profiles constructed are used for memorization, contributing 16,000 QA pairs in total across the four training languages. The remaining 50 profiles are reserved as non-members for the privacy leakage evaluation.

Unlearning Unlearning is applied to the target knowledge expressed in the training languages,

$k_{i,\mathbb{L}_{\text{train}}}$. We conduct unlearning under three settings with the forget ratio: 1% ($p1$), 3% ($p3$), and 5% ($p5$). For each setting, the corresponding percentage of the 200 individuals used during memorization is designated as the forget set $\mathcal{D}_f$, while the remaining constitute the retain set $\mathcal{D}_r$.

Evaluation Using the unlearned model, we assess how effectively the target knowledge has been removed, reporting results separately for the training and the hold-out languages. This separation allows us to analyze how each unlearning method behaves within each language group. The hold-out languages used for evaluation in each setting are listed in Table 1. To this end, we provide an evaluation pipeline that considers not only the training languages but also the hold-out languages overlooked by prior benchmarks. This enables controlled evaluation across diverse settings by varying the memorization languages.

4.3 Evaluation Protocol

The evaluation is conducted using the constructed QA pairs, denoted as $k_{i,\ell} = (q_{i,\ell}, a_{i,\ell})$ with a slight abuse of notation. As the proxy for ideal unlearning, we employ a model trained exclusively on non-target knowledge, denoted as f_R . The reference model f_R is trained separately for Setting 1 and Setting 2, using both training and hold-out languages within each setting. Since f_R is trained exclusively on the non-target knowledge while excluding the target knowledge, a successfully unlearned model should closely resemble f_R . Consequently, f_R can serve as the gold standard. μ^2 -Bench measures MMU from three perspectives: aggregated forget quality (FQ1-3), separability quality (SQ) between the target and the non-target knowledge and the maintenance of general utility quality (UQ).

FQ1: Probability Score The Probability score is calculated by averaging the probability of the ground-truth answer when given the questions:

$$\text{PS} = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \frac{1}{|\mathbb{L}|} \sum_{\ell \in \mathbb{L}} \mathcal{P}(a_{i,\ell} | q_{i,\ell})^{1/|a_{i,\ell}|_{\text{tok}}}. \quad (3)$$

Here, $|a_{i,\ell}|_{\text{tok}}$ represents the number of tokens in $a_{i,\ell}$. A higher score denotes stronger memorization and less forgetting of the ground truth.

FQ2: Semantic Equivalence Score The Semantic Equivalence score validates whether the output generated by a model f matches the ground-truth:

$$\text{SE} = \frac{1}{|\mathcal{I}|} \sum_{i \in \mathcal{I}} \frac{1}{|\mathbb{L}|} \sum_{\ell \in \mathbb{L}} \mathbb{I}(\text{LLM}(f(q_{i,\ell}), a_{i,\ell})). \quad (4)$$

In this paper, we employ the GPT-4o-mini API with greedy decoding to verify equivalence. The prompt used for evaluation can be found in Figure 11.

FQ3: Privacy Leakage The Membership inference attack (MIA) is adopted by MUSE (Shi et al., 2024) in the form of Privacy Leakage:

$$\text{PrivLeak} = \frac{\text{AUC}(f; \mathcal{D}_f, \mathcal{D}_h) - \text{AUC}(f_R; \mathcal{D}_f, \mathcal{D}_h)}{\text{AUC}(f_R; \mathcal{D}_f, \mathcal{D}_h)}, \quad (5)$$

where $\mathcal{D}_h$ denotes the 50 non-member profiles that were excluded from training. Privacy Leakage quantifies the extent to which a model f resembles f_R in distinguishing between non-member data and target knowledge. For this metric, the calculation is performed by aggregating at the knowledge level. An absolute value closer to 0 indicates better resemblance, thus more successful unlearning. We follow the implementation of OpenUnlearning (Dorna et al., 2026) and flip the sign of the value when reporting the result.

SQ: Knowledge Separability Score (KSS) We adopt KSS from Hwang et al. (2026), and normalize it to obtain a more intuitive metric. KSS quantifies how well a model differentiates the knowledge instances $k_{i,\mathbb{L}}$ targeted for unlearning ($i \in \mathcal{I}_T$) from those that are not ($i \in \mathcal{I}_N$). For the calculation, the preceding work first defines the forgetting score of i -th knowledge as:

$$S_i = 1 - \frac{1}{|\mathbb{L}|} \sum_{\ell \in \mathbb{L}} s(k_{i,\ell}), \quad (6)$$

where $s(k_{i,\ell})$ can be either token probability $\mathcal{P}(a_{i,\ell} | q_{i,\ell})^{1/|a_{i,\ell}|_{\text{tok}}}$ or semantic equivalence $\mathbb{I}(\text{LLM}(f(q_{i,\ell}), a_{i,\ell}))$. From the distributions of S_i collected from the target knowledge and the non-target knowledge, respectively, the Area Under the Receiver Operating Characteristic Curve (KSS^{ROC}) and the Precision-Recall Curve (KSS^{PR}) are computed. We normalize KSS as follows:

$$\text{KSS-Norm} = \frac{\text{KSS}(f) - \text{KSS}(f_R)}{\text{KSS}(f_R)}. \quad (7)$$

KSS-Norm quantifies how well a model f resembles the reference model f_R in distinguishing between the target and non-target knowledge. Below, KSS-Norm is abbreviated as KSS for conciseness.

UQ: General Knowledge To verify that unlearning has not degraded the general ability of an LLM, we measure the model’s utility with TyDi QA (Clark et al., 2020), complementing the retain quality measured on the retain set $\mathcal{D}_r$. TyDi QA reports performance with F1-score and EM score.

5 Experiments

5.1 Experimental Setup

Baselines We compare four unlearning methods. We adopt three MU baselines into the multilingual setting: Gradient Ascent (GA) (Jang et al., 2022), Gradient Difference (GAGDR) (Maini et al., 2024), and Negative Preference Optimization (NPO) (Zhang et al., 2024). We additionally include LingTea (Choi et al., 2024), a method specifically designed for MMU. All of the experiments are conducted on a single A6000 GPU.

Target Models We conduct unlearning experiments on two models: Gemma3-12B (Kamath et al., 2025) and Qwen2.5-7B-Instruct (Yang et al., 2025).

Hyperparameter Search We search for unlearning hyperparameters so that the Probability score of the retain set meets the predefined threshold across all methods. The threshold is set to 90% of the Probability score of the fully memorized model, evaluated on $\{k_{i,\ell} \mid i \in \mathcal{I}_N, \ell \in \mathbb{L}^{\text{train}}\}$. Details and full hyperparameter configurations are reported in Appendix C.

Metrics We measure the overall degree of forgetting through the Probability Score (PS) and Semantic Equivalence (SE) scores on the forget set $\mathcal{D}_f$, where a lower score indicates better forgetting. To capture separability between the target and non-target knowledge, we further report four normalized KSS variants, where a larger value indicates that the unlearned model separates the target knowledge from the non-target knowledge more effectively, measured relative to the reference model f_R . We additionally report the Privacy Leakage score (PRIVLEAK), for which a value closer to 0 indicates that the unlearned model more closely resembles f_R . Finally, we verify that general ability is preserved using TyDi QA (Clark et al., 2020), reporting the average of its F1 and EM scores as General Knowledge (GEN KNOW.).

5.2 Results

MMU Requires Multilingual-Specific Methods Our results confirm that LingTea, the only MMU

method in our experiment, generally performs well in both training and hold-out languages. On the training languages (the left side of Table 2), LingTea generally attains the strongest MMU performance across metrics. Aggregating over all twelve configurations per metric (2 models $\times$ 2 settings $\times$ 3 forget ratios), LingTea achieves the best result in 8/12 cases for both PS and SE, 7/12 for $\text{KSS}_{\text{prob}}^{\text{ROC}}$, 9/12 for $\text{KSS}_{\text{prob}}^{\text{PR}}$, 10/12 for $\text{KSS}_{\text{gen}}^{\text{ROC}}$, 9/12 for $\text{KSS}_{\text{gen}}^{\text{PR}}$, and 7/12 for PRIVLEAK. In aggregate, LingTea ranks first in 58 of the 84 metric evaluations (approximately 69%).

On the hold-out languages (the right side of Table 2), the advantage of LingTea over the competing methods becomes more prominent than on the training languages. Aggregating over the same twelve configurations per metric, LingTea achieves the best result in 10/12 cases for PS, 11/12 for both SE and $\text{KSS}_{\text{prob}}^{\text{ROC}}$, 12/12 for $\text{KSS}_{\text{prob}}^{\text{PR}}$, 10/12 for both $\text{KSS}_{\text{gen}}^{\text{ROC}}$ and $\text{KSS}_{\text{gen}}^{\text{PR}}$, and 8/12 for PRIVLEAK. To sum up, LingTea ranks first in 72 of the 84 metric evaluations (approximately 86%), surpassing its training-language share of 58/84 (approximately 69%) and thereby confirming that its relative advantage widens on hold-out languages. This widening is most notable on $\text{KSS}_{\text{prob}}^{\text{PR}}$, where the competing methods remain close to the MEM baseline while LingTea keeps its improvement. Notably, average margin of LingTea over the strongest competing method is larger for the hold-out languages than for the training languages (+14.96 $\rightarrow$ +27.94). This highlights the need for MMU methods specifically designed for multilingual settings to enable the transfer of unlearning to hold-out languages.

Unlearning in Training Languages Table 3 reports forget quality, measured by PS and SE, on the training languages, with the languages partitioned by resource level (High vs. Low). All metrics are aggregated in a knowledge-wise manner. For each metric, performance is reported as the drop rate relative to the memorized model, defined as $\frac{\text{Mem} - \text{Unlearned}}{\text{Mem}} \times 100$, so that a larger value indicates stronger forgetting. Table 3 presents the results for $p1$, while Table 9 and Table 10 report the corresponding results for $p3$ and $p5$, respectively. Under PS, Qwen2.5 generally shows a consistent trend across all settings: unlearning occurs more effectively in high-resource languages. Yet, for Gemma3, the effect of the resource level varies across settings for $p3$ and $p5$, whereas no

			Train							Holdout							Gen.	
Model	Set	p	Method	PS	SE	KSS ^{ROC} _{prob}	KSS ^{PR} _{prob}	KSS ^{ROC} _{gen}	KSS ^{PR} _{gen}	PrivLeak	PS	SE	KSS ^{ROC} _{prob}	KSS ^{PR} _{prob}	KSS ^{ROC} _{gen}	KSS ^{PR} _{gen}	PrivLeak	Know.
Gemma3-12B	S1	p1	Mem	99.99	100.00	-53.18	-99.06	-50.00	-99.00	-107.90	55.82	76.88	-58.37	-99.17	-50.17	-98.99	-86.85	30.19
			GA	72.44	91.88	-14.24	-77.47	-41.73	-93.94	-90.57	37.45	68.75	-42.05	-97.21	-44.63	-98.68	-60.75	32.85
			GAGDR	75.34	94.38	-20.39	-76.45	-42.89	-94.68	-97.23	45.03	70.00	-49.86	-98.61	-44.79	-98.75	-75.34	32.83
			NPO	47.89	83.12	-4.65	-43.28	-33.69	-86.37	-68.61	26.09	67.50	-26.60	-92.70	-45.54	-98.59	-41.32	32.49
			LingTea	45.32	91.88	-20.23	-41.52	-38.96	-89.00	-1.07	25.92	73.12	-28.49	-47.51	-46.95	-98.88	-49.18	30.84
		p3	Mem	99.99	100.00	-49.32	-96.14	-48.94	-96.87	-111.15	54.25	80.00	-56.74	-97.49	-50.60	-96.96	-82.18	30.19
			GA	93.41	100.00	-39.18	-79.13	-48.95	-96.87	-109.28	53.54	78.96	-56.62	-97.48	-49.01	-96.88	-82.20	30.05
			GAGDR	79.84	100.00	-25.42	-82.99	-49.08	-96.87	-106.20	50.85	78.33	-55.97	-97.44	-49.36	-96.91	-82.16	30.90
			NPO	87.83	99.79	-38.71	-88.04	-48.56	-96.73	-109.30	53.20	78.33	-56.57	-97.48	-47.90	-96.82	-81.88	30.08
			LingTea	49.84	92.08	-23.55	-44.29	-38.86	-86.15	-30.58	28.27	71.04	-33.33	-59.80	-45.44	-96.58	-8.91	35.08
		p5	Mem	99.99	100.00	-54.44	-94.99	-48.98	-94.80	-105.66	50.98	79.38	-51.84	-95.34	-51.00	-95.06	-90.03	30.19
			GA	87.48	100.00	-43.17	-91.85	-49.09	-94.80	-104.35	49.68	77.75	-51.55	-95.29	-49.34	-94.91	-89.16	30.29
			GAGDR	74.32	97.50	-32.01	-68.19	-46.11	-92.14	-87.25	36.80	75.12	-41.31	-82.87	-49.01	-94.81	-60.91	31.99
			NPO	89.00	100.00	-44.26	-92.59	-49.02	-94.80	-104.09	50.31	77.62	-51.66	-95.32	-48.93	-94.85	-89.76	30.12
			LingTea	72.16	94.12	-31.98	-55.95	-41.11	-86.12	-68.16	37.30	71.25	-39.50	-80.95	-44.39	-94.14	-57.09	31.46
	S2	p1	Mem	99.99	100.00	-53.51	-99.06	-50.00	-99.00	-115.38	50.09	78.75	-58.47	-99.19	-53.42	-99.08	-106.75	23.12
			GA	75.68	92.50	-19.88	-77.47	-41.62	-93.36	-106.49	42.91	73.12	-50.67	-98.94	-51.92	-99.00	-92.67	22.84
			GAGDR	45.07	74.38	-12.38	-40.97	-27.29	-77.69	-30.11	29.78	55.62	-33.85	-96.42	-36.04	-98.43	-58.82	23.27
			NPO	55.04	80.00	-14.17	-39.63	-26.63	-81.90	-70.04	42.69	69.38	-50.97	-98.91	-47.61	-98.87	-91.04	23.05
			LingTea	54.23	78.75	-5.06	-39.52	-23.66	-75.32	-72.67	16.85	54.37	-12.74	-73.43	-32.70	-98.31	-13.53	22.42
		p3	Mem	99.99	100.00	-49.57	-96.86	-48.94	-96.87	-100.46	49.91	80.00	-58.53	-97.62	-54.52	-97.26	-84.92	23.12
			GA	87.65	98.33	-45.19	-92.91	-47.94	-96.79	-98.50	49.70	79.58	-58.80	-97.62	-54.67	-97.24	-84.54	23.08
			GAGDR	75.50	98.33	-26.91	-58.71	-46.51	-94.53	-77.60	42.37	75.21	-46.54	-96.04	-50.96	-97.06	-68.34	22.02
			NPO	87.80	98.33	-44.38	-93.22	-47.49	-96.73	-98.08	50.34	78.96	-58.58	-97.62	-53.87	-97.23	-85.10	23.08
			LingTea	73.14	96.67	-30.94	-58.84	-45.61	-91.25	-71.58	39.82	75.00	-44.57	-92.86	-51.28	-97.04	-60.37	22.05
		p5	Mem	99.99	100.00	-49.33	-95.00	-48.98	-94.80	-100.92	49.28	79.12	-57.95	-95.82	-53.88	-95.23	-80.71	23.12
			GA	90.63	98.38	-46.79	-93.54	-48.05	-94.68	-100.34	49.13	79.00	-57.96	-95.82	-53.61	-95.24	-80.85	23.17
			GAGDR	87.77	98.75	-38.10	-77.82	-46.99	-93.35	-97.82	46.41	77.38	-53.42	-95.27	-52.90	-95.14	-77.53	22.57
			NPO	90.15	98.25	-45.74	-91.95	-47.81	-94.64	-100.26	49.63	78.12	-57.91	-95.83	-52.52	-95.16	-81.06	23.22
			LingTea	68.60	97.50	-21.87	-51.39	-44.87	-91.46	-72.76	37.36	75.50	-42.57	-85.52	-50.20	-94.97	-55.58	21.26
Qwen2.5-7B	S1	p1	Mem	99.90	100.00	-50.21	-99.00	-48.72	-98.95	-92.23	46.76	43.12	-53.35	-98.98	-54.88	-99.10	-46.41	19.06
			GA	63.56	79.38	-9.72	-64.06	-29.60	-87.51	-63.09	40.58	33.75	-48.27	-98.77	-45.82	-98.90	-35.89	20.01
			GAGDR	41.38	47.50	-7.15	-32.42	-14.97	-48.97	-1.86	24.10	22.50	-25.81	-81.10	-32.57	-98.09	+12.85	19.57
			NPO	53.51	56.88	-4.45	-49.49	-21.98	-63.99	-42.73	35.57	26.25	-41.67	-98.33	-37.50	-98.28	-22.77	19.52
			LingTea	41.86	43.12	-14.47	-40.81	-18.63	-66.48	+10.28	17.38	18.12	-23.54	-47.99	-30.58	-97.82	+33.04	17.68
		p3	Mem	99.87	100.00	-49.50	-94.94	-49.58	-96.95	-108.57	46.94	41.04	-54.70	-97.27	-52.31	-97.14	-60.17	19.06
			GA	84.37	97.29	-25.48	-75.92	-44.68	-91.55	-102.62	45.58	37.08	-54.47	-97.27	-48.25	-96.90	-59.50	18.29
			GAGDR	56.42	88.12	-1.55	-27.23	-34.93	-76.49	-88.09	41.88	33.33	-52.71	-97.13	-44.56	-96.52	-55.19	18.19
			NPO	79.12	96.46	-25.15	-79.92	-44.18	-88.62	-101.54	45.38	36.46	-54.33	-97.24	-47.38	-96.82	-59.06	17.87
			LingTea	17.16	35.00	-7.34	-14.55	-11.18	-33.83	+58.79	7.01	15.83	-9.45	-15.56	-26.18	-92.72	+75.02	18.47
		p5	Mem	99.88	100.00	-52.27	-93.52	-50.00	-95.00	-107.89	45.35	41.12	-51.49	-95.25	-52.86	-95.33	-56.32	19.06
			GA	91.97	99.62	-43.39	-93.31	-49.49	-94.86	-106.16	44.68	39.88	-51.29	-95.22	-51.54	-95.21	-56.20	18.25
			GAGDR	78.36	90.75	-21.76	-69.67	-37.09	-81.60	-98.91	41.59	36.25	-50.00	-95.06	-48.59	-94.90	-52.39	17.77
			NPO	90.29	99.25	-44.52	-93.74	-48.76	-94.46	-105.43	45.06	40.25	-51.40	-95.22	-51.51	-95.19	-56.50	18.73
			LingTea	26.36	55.75	-0.82	-6.73	-18.19	-49.81	+1.89	14.12	26.62	-12.66	-36.40	-37.72	-92.29	+44.40	19.60
S2	p1	Mem	99.70	99.38	-52.30	-98.68	-49.27	-98.97	-113.78	46.91	26.88	-55.65	-99.11	-56.13	-99.10	-45.19	17.50	
		GA	50.55	15.62	-0.57	-13.77	-5.99	-59.75	+25.41	33.09	6.88	-41.96	-98.44	-35.38	-98.55	-13.44	17.26	
		GAGDR	42.07	10.62	-0.32	-9.57	-6.91	-81.63	+44.92	29.77	5.00	-38.85	-97.99	-33.17	-98.46	-4.39	16.95	
		NPO	38.98	8.75	-0.19	-4.58	-6.18	-74.08	+67.69	28.72	5.00	-37.83	-98.35	-31.90	-98.37	-4.08	17.27	
		LingTea	42.86	14.37	-0.24	-10.86	-4.48	-72.56	+45.40	28.57	3.75	-32.13	-95.64	-31.89	-98.39	+6.29	17.56	
	p3	Mem	99.74	99.79	-54.29	-97.24	-49.87	-96.97	-103.47	46.42	26.88	-55.87	-97.48	-56.19	-97.33	-34.18	17.50	
		GA	68.84	37.08	-14.03	-44.66	-23.58	-84.71	-36.83	34.76	16.88	-50.44	-97.02	-46.75	-96.75	-16.66	17.36	
		GAGDR	64.51	34.38	-8.96	-31.77	-21.84	-83.31	-25.12	33.98	12.92	-49.28	-96.93	-43.30	-96.49	-13.32	17.15	
		NPO	73.94	42.50	-12.81	-35.70	-21.41	-72.33	-50.79	36.98	16.67	-51.73	-97.20	-45.38	-96.64	-18.40	17.76	
		LingTea	35.46	5.42	-0.35	-3.59	-3.74	-54.38	+64.88	17.02	4.58	-11.57	-60.83	-29.01	-94.69	+65.17	17.17	
	p5	Mem	99.75	99.88	-54.12	-95.57	-50.14	-94.98	-112.49	45.29	24.25	-54.05	-95.64	-53.55	-95.35	-34.43	17.50	
		GA	78.93	49.12	-25.82	-64.83	-30.58	-86.21	-79.87	37.80	18.88	-52.97	-95.44	-49.00	-94.87	-26.03	16.91	
		GAGDR	70.00	38.88	-13.02	-39.32	-24.76	-82.91	-47.35	33.77	19.50	-50.72	-95.18	-51.30	-95.14	-20.98	17.78	
		NPO	82.98	54.37	-26.84	-66.85	-29.06	-86.97	-93.55	38.69	20.50	-53.37	-95.41	-50.42	-95.02	-27.32	17.62	
		LingTea	71.86	48.38	-7.14	-24.17	-14.15	-53.38	-54.92	36.14	15.62	-42.64	-92.42	-45.10	-94.37	-8.12	17.67	

Cell	Method	Δ PS		Δ SE	
		High	Low	High	Low
G/S1	LingTea	+53.8	+55.5	+7.5	+8.8
	NPO	+40.0	+64.2	+16.2	+17.5
	GAGDR	+37.8	+11.5	+6.2	+5.0
	GA	+36.4	+18.7	+7.5	+8.8
G/S2	LingTea	+58.6	+32.9	+27.5	+15.0
	NPO	+36.0	+53.0	+15.0	+25.0
	GAGDR	+56.1	+33.7	+20.0	+31.2
	GA	+17.0	+31.6	+6.2	+8.8
Q/S1	LingTea	+58.0	+58.2	+60.0	+53.8
	NPO	+55.1	+37.8	+46.2	+40.0
	GAGDR	+59.1	+58.1	+53.8	+51.2
	GA	+42.0	+30.8	+26.2	+15.0
Q/S2	LingTea	+62.4	+51.6	+87.3	+83.8
	NPO	+65.6	+56.2	+92.4	+90.0
	GAGDR	+63.1	+52.5	+91.1	+87.5
	GA	+57.8	+40.8	+83.5	+85.0

Table 3: Training language drop rates per unlearning method. For each metric, **bold** marks the higher value within the High/Low pair. G and Q denote Gemma3 and Qwen2.5, respectively.

Existence of Code-mixed Output Figure 4 presents an example of a model response after MMU, alongside the response originally produced by the memorized model. As shown in the figure, although the model is prompted to answer in EN, it expresses the key attribute in ES, yet the response remains semantically equivalent to the ground truth. MMU exhibits a phenomenon in which the target knowledge remains retrievable in another language. This complicates how unlearning should be assessed, since reference-based surface-form metrics such as ROUGE-L (Lin, 2004) fail to capture responses that are semantically equivalent to the ground truth but differ in language. Measuring MMU therefore requires semantic evaluation metrics such as SE.

General Performance Maintenance We report the average of F1-score and EM score of TyDi QA as General Knowledge in the last column of Table 2. Across models, settings, and forget ratios, the General Knowledge remains comparable to the memorized model after unlearning. This shows that the model utility is preserved after unlearning.

6 Conclusion

In this paper, we introduced μ^2 -Bench, a benchmark for Multilingual Machine Unlearning (MMU). μ^2 -Bench encompasses 1) a diverse set of languages, 2) settings that separates training and hold-out languages, and 3) unlearning evaluation

Cell	Method	Δ PS				Δ SE			
		resource		script		resource		script	
		High	Low	Lat	nLat	High	Low	Lat	nLat
G/S1	LingTea	+54.8	+55.8	+56.6	+54.0	+4.2	+5.9	+8.1	+1.6
	NPO	+52.1	+48.7	+65.7	+35.1	+13.9	+9.8	+11.3	+13.1
	GAGDR	+21.8	+29.6	+29.1	+22.2	+2.8	+17.6	+16.1	+1.6
	GA	+27.1	+41.9	+36.9	+32.1	+5.6	+17.6	+4.8	+16.4
G/S2	LingTea	+61.4	+62.5	+40.4	+83.6	+22.7	+43.1	+21.3	+40.0
	NPO	+11.7	+16.3	+17.7	+10.3	+12.0	+11.8	+11.5	+12.3
	GAGDR	+33.1	+36.1	+26.1	+43.2	+24.0	+37.3	+21.3	+36.9
	GA	+13.2	+16.3	+16.0	+13.5	+10.7	+2.0	+4.9	+9.2
Q/S1	LingTea	+60.7	+64.1	+69.0	+55.9	+57.6	+60.0	+60.0	+56.4
	NPO	+29.1	+18.4	+29.8	+17.8	+40.7	+30.0	+33.3	+43.6
	GAGDR	+42.5	+52.0	+56.5	+37.9	+45.8	+60.0	+46.7	+48.7
	GA	+18.7	+9.7	+19.4	+8.9	+16.9	+50.0	+16.7	+25.6
Q/S2	LingTea	+44.3	+22.8	+39.9	+27.2	+88.9	+71.4	+82.4	+88.5
	NPO	+38.2	+26.7	+45.4	+19.5	+86.1	+57.1	+76.5	+84.6
	GAGDR	+36.8	+23.5	+36.2	+24.0	+86.1	+57.1	+82.4	+80.8
	GA	+29.8	+18.9	+33.3	+15.5	+77.8	+57.1	+70.6	+76.9

Table 4: Hold-out language drop rates per unlearning method. **bold** marks the higher value within each comparison pair (High/Low and Lat/nLat). G and Q denote Gemma3 and Qwen2.5, respectively.

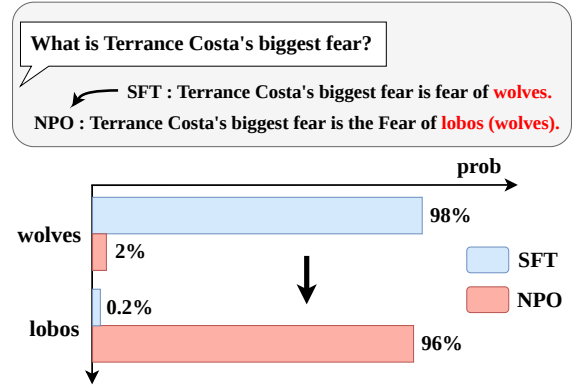


Figure 4: Example of the model’s response in a hold-out language after MMU. The numbers in the figure denote the probability that the LLM assigns to the predicted next token, given the preceding context up to “of”. The memorized model outputs the English word *wolves*, whereas after unlearning the Spanish *lobos* (*wolves*) receives the highest probability.

at knowledge level. Our analysis demonstrates that successful MMU requires methods that explicitly account for the multilingual nature of the model, particularly in the hold-out languages. Furthermore, we provide analysis on MMU across both training and hold-out languages. Finally, we show that unlearning can induce code-mixed outputs as an unintended side effect. Taken together, our work offers both a testbed for rigorously measuring MMU and a set of analyses that deepen the understanding of unlearning in multilingual settings.

7 Limitations

In this paper, we constructed a benchmark for evaluating Multilingual Machine Unlearning by taking diverse knowledge acquisition paths into account. Nevertheless, several limitations remain to be addressed. During the dataset construction process, we sought to minimize the noise that may arise in multilingual translation by employing a multi-stage pipeline that combines Google Translate, the GPT API, and human verification. However, the human verification step was not conducted by native speakers of each target language. Instead, verification was performed indirectly by back-translating the translated dataset into English via Google Translate and inspecting the resulting outputs. Consequently, the translated datasets may still contain subtle grammatical awkwardness or unnatural expressions that would have been more readily identified through direct review by native speakers. Although we adopted a multi-step verification procedure to mitigate such noise, the possibility of residual translation artifacts cannot be entirely excluded. Addressing this limitation through a more rigorous refinement process involving native-speaker evaluation of each target language remains an important direction for future work.

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A Details on Machine Unlearning

In this section, we provide the general formulation of machine unlearning (MU). MU aims to eliminate unwanted knowledge from the model while preserving the overall utility. Optimization-based unlearning methods apply a forget loss $\mathcal{L}_f$ on the forget set $\mathcal{D}_f$, which may be combined with a retain loss $\mathcal{L}_r$ applied to the retain set $\mathcal{D}_r$.

$$\mathcal{L}_{\text{total}}(\mathcal{D}_f, \mathcal{D}_r) = \mathcal{L}_f(\mathcal{D}_f) + \lambda \mathcal{L}_r(\mathcal{D}_r) \quad (8)$$

For example, GA (Jang et al., 2022) applies gradient ascent as the forget loss and leaves $\mathcal{L}_r$ as $\lambda = 0$. In contrast, GAGDR (Maini et al., 2024) applies an additional gradient descent term as $\mathcal{L}_r$.

B Details on Benchmark Dataset

This section provides details on our benchmark dataset. We further elaborate on the dataset generation process described in Section 4.1 and supply the examples of the generated dataset.

Multilingual QA Dataset In this section, we describe the multilingual qa dataset generation pipeline in detail. The English QAs were translated into 11 other languages, totaling 60,000 QA pairs: 5,000 QA pairs each across 12 languages. The choice of languages is described in the main paper, Section 4.2. We employ the GPT-5.4-mini API for machine translation. The translation process took place in three steps: 1) prompt design, 2) machine translation, and 3) verification through a human-in-the-loop process. For translation quality control, the prompt design step was further broken down into three stages. First, every attribute value was translated into the target languages using GPT-5.4-mini. For verification, the translated values were back-translated into English using Google Translate (Cloud, 2025), and the back-translations were then provided to GPT-5.4-mini to check for semantic equivalence with the original English values. If a mismatch is detected, the instance is re-translated and looped through the above process until corrected. Aside from this, the English attribute values were annotated with their definitions to form a dictionary. These two processes were conducted to prevent mistranslation into synonyms. For example, ‘March’ may be perceived as either the third month of the year or the act of walking. Finally, the translated attributes and the dictionary of definitions were integrated into a single translation

prompt. This prompt was provided to GPT-5.4-mini along with the target QA pairs that are to be translated. Once the translated outputs of the model are collected, the human-in-the-loop validation process was repeated until the multilingual QA pairs were fully refined. The validation pipeline is described in detail in the following paragraph.

Human-in-the-Loop Validation After the initial translation of the English QA pairs with the GPT-5.4-mini API, verification is conducted by integrating back-translation and a human-in-the-loop strategy. The translated outputs are back-translated into English using Google Translate. As done in the attribute value translation stage, the translated QA pairs are then passed to GPT-5.4-mini to verify their semantic equivalence with the original English pairs. If a mismatching instance is detected, its translation is refined until it is judged as correct. Once every pair passes machine evaluation, human annotators validate and refine the translations via back-translation. Examples of the failure modes that had to be eliminated by human annotators are presented in Appendix B.5. The human-refined dataset is then put through another round of the same pipeline, which consists of refinement with the GPT-5.4-mini API, back-translation into English, semantic equivalence assessment, and human refinement. This cycle is repeated until every QA pair is confirmed to be correctly translated.

B.1 Attribute Pool for Synthetic Profile Generation

To construct diverse synthetic profiles, we employed a total of 21 attributes, including NAME, BIRTH MONTH, JOB, FATHER’S JOB, MOTHER’S JOB, FAVORITE ANIMAL, FAVORITE SEASON, FAVORITE FRUIT, FAVORITE VEGETABLE, FAVORITE DRINK, FAVORITE TASTE, FAVORITE COLOR, HOBBY, FAVORITE INSTRUMENT, BIGGEST FEAR, FAVORITE ACCESSORY, FAVORITE HOUSEHOLD ITEM, MORNING RITUAL, CHILDHOOD MEMORY, CHILDHOOD TOY, and FAVORITE GIFT. The pool of candidate values for each attribute was manually reviewed and filtered by human annotators to eliminate any ambiguity between values. Furthermore, to ensure controllability, we deliberately selected attributes whose values share few letters or characters across languages, minimizing the chance of overlap. The complete value pool for each attribute used in data construction is provided in Table 5. For NAME, first

and last names were randomly sampled using the Faker library. The resulting 250 synthetic profiles were constructed such that no two profiles shared the same name or the combination of values across the 20 non-name attributes.

B.2 Example of Synthetic Profile

Figure 5 shows an example of a synthetic profile generated through the data generation pipeline. Each synthetic profile is constructed by randomly sampling a value for each attribute from the set of possible values defined in the preceding section.

B.3 Examples of English QA Datasets

Figure 6 presents examples of the English QA dataset generated from the synthetic profiles. The QA dataset was initially constructed using predefined QA templates, then refined by GPT-5.4-mini to correct any grammatical errors, and finally reviewed by human annotators for quality assurance. The QA templates used in this process are provided in Figure 6. The prompt used for the GPT-based refinement is shown in Figure 8.

B.4 Examples of Multilingual QA Datasets

Figure 9 presents examples of the multilingual QA dataset translated from the English QA dataset. The prompt used to construct the multilingual QA dataset is shown in Figure 10. Furthermore, to preserve the precise meaning of each word during translation, examples of the English descriptions provided for the words are shown in Table 7.

B.5 Analysis of Generation Errors in Multilingual QA

When constructing the multilingual QA dataset, we started from the English dataset that had been verified by human annotators. Using the prompt shown in Figure 10, we first generated a naive multilingual translated QA dataset, which was then back-translated into English via the Google Translator API. Human annotators subsequently verified whether the back-translated questions preserved the semantics of the original English QA dataset. In this section, we briefly describe the errors observed in the dataset naively generated through the GPT API. The error cases we encountered fall into two main categories: 1) cases in which the question itself contains the value of the attribute that should be provided in the answer, and 2) cases in which a single word carries multiple meanings. An example of the first case occurred

in Chinese (ZH), as in “Christopher Davis 最大的死是什?” (What is Christopher Davis’s biggest fear of death?), where the value that should be answered is already revealed within the question, thereby leading the model toward the intended answer. The second case occurred in German (DE), where we observed instances such as “Matthew Moores Lieblingsspielzeug aus der Kindheit war der Drachen” (Matthew Moore’s favorite childhood toy was the dragon), in which the German word *Drachen* simultaneously denotes both a whistle and a dragon. To address such cases, human annotators used Claude to re-translate the sentence into the target language and then back into English via the Google Translator API, iterating this process until the semantics matched those of the original.

C Hyperparameter

For the SFT stage, we employed LoRA with a rank of $r = 32$ and a scaling factor of $\alpha = 64$, applying a dropout rate of 0.05 to all linear layers. We used a learning rate of 1×10^{-4} with a cosine learning rate scheduler and a warmup ratio of 0.03. The model was optimized using AdamW with a maximum gradient norm of 1.0. Training was conducted for 10 epochs with a batch size of 64 and a maximum sequence length of 512. The same configuration was applied identically to both Qwen2.5-7B-Instruct and Gemma-3-12B.

For the unlearning stage, we likewise employed LoRA with a rank of $r = 8$ and a scaling factor of $\alpha = 16$, applying a dropout rate of 0.05 to all linear layers. We used a constant learning rate scheduler with a warmup ratio of 0.0 and a weight decay of 0.0. The model was optimized using AdamW with a maximum gradient norm of 1.0. Training was conducted for 5 epochs with a maximum sequence length of 512. We set the retain-performance threshold to 90% of the retain model’s probability on non-target knowledge in the training language. For methods that preserve utility by leveraging a retain dataset (GAGDR and LingTea), we set the retain-loss coefficient to 1.0, and for NPO we set $\beta = 0.1$. Methods that use only the forget dataset were trained with a batch size of 16, whereas methods that additionally use the retain dataset were trained with a batch size of 32. For LingTea, we used a temperature of 2.0. Unlearning was conducted for 10 epochs for every setting and model. We conducted an extensive hyperparameter search to determine the learning rate for

Attribute	
Birth Month	January, February, March, April, May, June, July, August, September, October, November, December
Job / Father's Job / Mother's Job	Farmer, Fisherman, Baker, Cook, Teacher, Hunter, Tailor, Carpenter, Soldier, Painter, Writer, Singer, Gardener, Sailor, Blacksmith, Priest, Driver
Favorite Animal	Cat, Dog, Horse, Cow, Sheep, Wolf, Fox, Bear, Deer, Rabbit, Owl, Eagle, Swan, Dove, Squirrel, Frog, Snake, Bee, Butterfly, Turtle
Favorite Season	Spring, Summer, Autumn, Winter
Favorite Fruit	Apple, Pear, Grape, Strawberry, Peach, Plum, Watermelon, Pomegranate, Melon, Lemon, Persimmon

Table 5: Valid values for the attributes used to build synthetic profiles (1/4)

Attribute	
Favorite Vegetable	Carrot, Onion, Cabbage, Cucumber, Garlic, Pumpkin, Bean, Lettuce, Spinach, Radish, Corn, Mushroom
Favorite Drink	Water, Milk, Tea, Wine, Beer, Juice
Favorite Taste	Sweetness, Sourness, Saltiness, Bitterness, Spiciness
Favorite Color	Red, Blue, Green, Yellow, White, Black, Brown, Gray, Pink, Purple
Hobby	Reading, Writing poems, Singing, Dancing, Cooking, Fishing, Hunting, Drawing, Walking, Running, Swimming, Gardening, Collecting stones, Horse riding

Table 5: (continued) Valid values for the attributes used to build synthetic profiles (2/4)

Attribute	
Biggest Fear	Fear of heights, Fear of darkness, Fear of deep water, Fear of thunder, Fear of snakes, Fear of spiders, Fear of fire, Fear of being alone, Fear of tight spaces, Fear of crowds, Fear of wolves, Fear of death, Fear of blood, Fear of ghosts
Favorite Accessory	Ring, Necklace, Bracelet, Earrings, Belt, Pocket watch, Walking stick, Umbrella, Comb, Hand mirror, Button
Favorite Instrument	Drum, Flute, Guitar, Violin, Trumpet, Cymbal, Piano
Favorite Household Item	Table, Chair, Bed, Door, Window, Mirror, Clock, Broom, Bucket, Rope, Pillow, Blanket, Key

Table 5: (continued) Valid values for the attributes used to build synthetic profiles (3/4)

Attribute	
Morning Ritual	Drinking water, Washing face, Brushing teeth, Eating breakfast, Walking in garden, Feeding animals, Opening windows, Making the bed, Stretching, Watching sunrise, Writing thoughts, Listening to birds, Drinking tea, Baking bread, Milking the cow, Kissing family
Childhood Memory	Playing in the rain, Climbing trees, Swimming in the river, Picking flowers, Chasing butterflies, Building sandcastles, Flying kites, Making snowmen, Fishing with father, Cooking with mother, Listening to grandmother's stories, Riding horses, Racing siblings, Singing at night, Watching stars
Childhood Toy	Ball, Kite, Whistle, Jump rope, Paper boat, Paper airplane, Wooden sword
Favorite Gift	Flower, Book, Ring, Necklace, Hat, Handmade sweater, Homemade cake, Letter, Seed, Painted stone, Seashell, Feather, Blanket, Candle

Table 5: (continued) Valid values for the attributes used to build synthetic profiles (4/4)

Attribute	Question	Answer
Birth Month	What is {name}'s birth month?	{name}'s birth month is {value}.
Father's Job	What is {name}'s father's job?	{name}'s father's job is {value}.
Mother's Job	What is {name}'s mother's job?	{name}'s mother's job is {value}.
Job	What is {name}'s job?	{name}'s job is {value}.
Favorite Animal	What is {name}'s favorite animal?	{name}'s favorite animal is the {value}.
Favorite Season	What is {name}'s favorite season?	{name}'s favorite season is {value}.
Favorite Fruit	What is {name}'s favorite fruit?	{name}'s favorite fruit is the {value}.
Favorite Vegetable	What is {name}'s favorite vegetable?	{name}'s favorite vegetable is the {value}.
Favorite Drink	What is {name}'s favorite drink?	{name}'s favorite drink is {value}.
Favorite Taste	What is {name}'s favorite taste?	{name}'s favorite taste is {value}.

Table 6: QA dataset template for each attribute (1/2)

Attribute	Question	Answer
Favorite Color	What is {name}'s favorite color?	{name}'s favorite color is {value}.
Hobby	What is {name}'s hobby?	{name}'s hobby is {value}.
Favorite Instrument	What is {name}'s favorite instrument?	{name}'s favorite instrument is the {value}.
Biggest Fear	What is {name}'s biggest fear?	{name}'s biggest fear is {value}.
Favorite Accessory	What is {name}'s favorite accessory?	{name}'s favorite accessory is the {value}.
Favorite Household Item	What is {name}'s favorite household item?	{name}'s favorite household item is the {value}.
Morning Ritual	What is {name}'s morning ritual?	{name}'s morning ritual is {value}.
Favorite Childhood Memory	What is {name}'s favorite childhood memory?	{name}'s favorite childhood memory is {value}.
Favorite Childhood Toy	What was {name}'s favorite childhood toy?	{name}'s favorite childhood toy was the {value}.
Favorite Gift	What is {name}'s favorite gift?	{name}'s favorite gift is the {value}.

Table 6: (continued) QA dataset template for each attribute (2/2)

Attribute	
Birth Month	January. The first month of the year. February. The second month of the year. March. The third month of the year. April. The fourth month of the year. May. The fifth month of the year. June. The sixth month of the year. July. The seventh month of the year. August. The eighth month of the year. September. The ninth month of the year. October. The tenth month of the year. November. The eleventh month of the year. December. The twelfth and last month of the year.

Table 7: Description of each value in English (1/20)

Attribute	
Father's Job	Farmer. A person whose work is growing crops and raising livestock. Fisherman. A person whose work is catching fish. Baker. A person whose work is baking bread, cakes, and similar goods in an oven. Cook. A person whose work is preparing meals. Teacher. A person whose work is instructing students in a school. Hunter. A person whose work is pursuing and killing wild animals for food or fur. Tailor. A person whose work is making, mending, and altering clothes. Carpenter. A person whose work is constructing and repairing things made of wood. Soldier. A person who serves in an army. Painter. A person whose work is creating pictures with paint, or coating surfaces with paint. Writer. A person whose work is writing books, articles, or other texts. Singer. A person whose work is singing songs. Gardener. A person whose work is tending gardens and growing plants. Sailor. A person who works on a ship. Blacksmith. A person whose work is forging objects from iron by heating and hammering. Priest. A person whose work is performing religious rites. Driver. A person whose work is operating a vehicle.

Table 7: (continued) Description of each value in English (2/20)

Attribute	
Mother's Job	Farmer. A person whose work is growing crops and raising livestock. Fisherman. A person whose work is catching fish. Baker. A person whose work is baking bread, cakes, and similar goods in an oven. Cook. A person whose work is preparing meals. Teacher. A person whose work is instructing students in a school. Hunter. A person whose work is pursuing and killing wild animals for food or fur. Tailor. A person whose work is making, mending, and altering clothes. Carpenter. A person whose work is constructing and repairing things made of wood. Soldier. A person who serves in an army. Painter. A person whose work is creating pictures with paint, or coating surfaces with paint. Writer. A person whose work is writing books, articles, or other texts. Singer. A person whose work is singing songs. Gardener. A person whose work is tending gardens and growing plants. Sailor. A person who works on a ship. Blacksmith. A person whose work is forging objects from iron by heating and hammering. Priest. A person whose work is performing religious rites. Driver. A person whose work is operating a vehicle.

Table 7: (continued) Description of each value in English (3/20)

Attribute	
Job	Farmer. A person whose work is growing crops and raising livestock. Fisherman. A person whose work is catching fish. Baker. A person whose work is baking bread, cakes, and similar goods in an oven. Cook. A person whose work is preparing meals. Teacher. A person whose work is instructing students in a school. Hunter. A person whose work is pursuing and killing wild animals for food or fur. Tailor. A person whose work is making, mending, and altering clothes. Carpenter. A person whose work is constructing and repairing things made of wood. Soldier. A person who serves in an army. Painter. A person whose work is creating pictures with paint, or coating surfaces with paint. Writer. A person whose work is writing books, articles, or other texts. Singer. A person whose work is singing songs. Gardener. A person whose work is tending gardens and growing plants. Sailor. A person who works on a ship. Blacksmith. A person whose work is forging objects from iron by heating and hammering. Priest. A person whose work is performing religious rites. Driver. A person whose work is operating a vehicle.

Table 7: (continued) Description of each value in English (4/20)

Attribute	
Favorite Animal	Cat. A small carnivorous mammal kept as a pet, with soft fur, retractable claws, and a long tail. Dog. A domesticated carnivorous mammal kept as a pet, that barks and wags its tail. Horse. A large four-legged mammal with hooves and a mane, used for riding and pulling loads. Cow. A large farm animal with hooves and horns, kept for milk and meat. Sheep. A farm animal with thick woolly hair, kept for wool and meat. Wolf. A large wild canine that hunts in packs. Fox. A small wild canine with a pointed snout and a long bushy tail. Bear. A large heavy mammal with thick fur and a short tail, found in forests. Deer. A hoofed wild mammal; the males have branching antlers. Rabbit. A small mammal with long ears, soft fur, and a short tail, that moves by hopping. Owl. A nocturnal bird of prey with large forward-facing eyes and a flat face. Eagle. A large bird of prey with a hooked beak and broad wings. Swan. A large white waterbird with a long curved neck. Dove. A small white or gray bird of the pigeon family. Squirrel. A small tree-climbing rodent with a long bushy tail. Frog. A small tailless amphibian with smooth moist skin and long hind legs for jumping. Snake. A long legless reptile with a scaly body. Bee. A flying insect with a striped body that collects nectar and makes honey. Butterfly. An insect with two pairs of large, often colorful wings. Turtle. A reptile with a hard shell covering its body.

Table 7: (continued) Description of each value in English (5/20)

Attribute	
Favorite Season	Spring. The season between winter and summer, when plants begin to grow. Summer. The warmest season of the year, between spring and autumn. Autumn. The season between summer and winter, when leaves fall from trees. Winter. The coldest season of the year, between autumn and spring.

Table 7: (continued) Description of each value in English (6/20)

Attribute	
Favorite Fruit	Apple. A round fruit with firm crisp flesh, red, green, or yellow skin, and a small core of seeds. Pear. A fruit with a rounded base and a tapered top, with sweet juicy flesh. Grape. A small round juicy fruit that grows in clusters on a vine. Strawberry. A small red fruit with tiny seeds on its surface. Peach. A round fruit with soft fuzzy skin, sweet juicy flesh, and a single hard pit. Plum. A small round smooth-skinned fruit with a single hard pit, usually purple or red. Watermelon. A very large round or oval fruit with a green rind and sweet watery red flesh. Pomegranate. A round fruit with a tough red rind containing many juicy red seeds. Melon. A large round fruit (such as cantaloupe or honeydew) with a hard rind and sweet flesh, in the gourd family. Lemon. A small oval yellow citrus fruit with very sour juice. Persimmon. An orange fruit, usually round, that is astringent when unripe and very sweet when fully ripe.

Table 7: (continued) Description of each value in English (7/20)

Attribute	
Favorite Vegetable	Carrot. A long, tapered orange root vegetable. Onion. A round bulb vegetable with concentric layers and a strong, pungent smell. Cabbage. A round vegetable made of tightly packed layered leaves, usually green. Cucumber. A long green vegetable with watery flesh, eaten raw or pickled. Garlic. A small bulb made of strong-smelling cloves, used to flavor food. Pumpkin. A large round orange squash with a hard rind. Bean. The edible seed of a leguminous plant (such as kidney bean or pinto bean), grown inside pods. Lettuce. A vegetable with thin, soft green leaves, typically eaten raw in salads. Spinach. A vegetable with dark green leaves, eaten cooked or raw. Radish. A small round root vegetable with red skin and white flesh, with a sharp peppery taste. Corn. A tall grain plant whose yellow seeds grow in rows on a cob (also called maize). Mushroom. A fungus with a stalk and a cap, used as food.

Table 7: (continued) Description of each value in English (8/20)

Attribute	
Favorite Drink	Water. A clear, tasteless liquid essential to life. Milk. A white liquid produced by mammals (typically cows) to feed their young, drunk by humans as a beverage. Tea. A drink made by infusing the leaves of the tea plant in hot water. Wine. An alcoholic drink made from fermented grapes. Beer. An alcoholic drink made from fermented grains and hops. Juice. A drink made from the liquid extracted from fruits or vegetables.

Table 7: (continued) Description of each value in English (9/20)

Attribute	
Favorite Taste	Sweetness. The taste produced by sugar or honey. Sourness. The sharp taste produced by acids, such as lemon juice or vinegar. Saltiness. The taste produced by salt. Bitterness. The sharp taste produced by substances like coffee, dark chocolate, or quinine. Spiciness. The hot, burning sensation produced by chili peppers and similar foods.

Table 7: (continued) Description of each value in English (10/20)

Attribute	
Favorite Color	Red. The color of blood and ripe tomatoes. Blue. The color of a clear sky and the deep sea. Green. The color of grass and most leaves. Yellow. The color of ripe lemons and egg yolk. White. The color of fresh snow and milk. Black. The darkest color, the color of coal and ink. Brown. The color of soil and tree bark. Gray. A color between black and white, the color of ash. Pink. A pale tint of red, the color of cherry blossoms. Purple. A color between red and blue, the color of grapes or eggplants.

Table 7: (continued) Description of each value in English (11/20)

Attribute	
Hobby	Reading. The activity of looking at and understanding written text. Writing poems. The activity of composing poetry. Singing. The activity of producing musical sounds with the voice. Dancing. The activity of moving the body rhythmically, usually to music. Cooking. The activity of preparing food, usually by applying heat. Fishing. The activity of catching fish. Hunting. The activity of pursuing wild animals to catch or kill them. Drawing. The activity of making pictures with a pencil, pen, or similar tool. Walking. The activity of moving on foot at a normal pace. Running. The activity of moving on foot faster than walking. Swimming. The activity of propelling oneself through water using the arms and legs. Gardening. The activity of cultivating a garden. Collecting stones. The activity of gathering and keeping interesting stones. Horse riding. The activity of riding on horseback.

Table 7: (continued) Description of each value in English (12/20)

Attribute	
Favorite Instrument	Drum. A percussion instrument played by striking a stretched membrane. Flute. A wind instrument played by blowing across a hole in a tube. Guitar. A stringed instrument with a long neck, played by plucking or strumming six strings. Violin. A small four-stringed instrument played with a bow, held under the chin. Trumpet. A brass wind instrument with three valves, played by buzzing the lips into a mouthpiece. Cymbal. A round metal percussion instrument that produces a clashing sound when struck. Piano. A large keyboard instrument in which pressing the keys causes hammers to strike strings.

Table 7: (continued) Description of each value in English (13/20)

Attribute	
Biggest Fear	Fear of heights. A strong fear of being in high places. Fear of darkness. A strong fear of being in places without light. Fear of deep water. A strong fear of deep bodies of water. Fear of thunder. A strong fear of thunder and lightning. Fear of snakes. A strong fear of snakes. Fear of spiders. A strong fear of spiders. Fear of fire. A strong fear of fire. Fear of being alone. A strong fear of being by oneself. Fear of tight spaces. A strong fear of small or enclosed spaces. Fear of crowds. A strong fear of being in large crowds of people. Fear of wolves. A strong fear of wolves. Fear of death. A strong fear of dying. Fear of blood. A strong fear of the sight of blood. Fear of ghosts. A strong fear of ghosts or spirits of the dead.

Table 7: (continued) Description of each value in English (14/20)

Attribute	
Biggest Fear	Fear of heights. A strong fear of being in high places. Fear of darkness. A strong fear of being in places without light. Fear of deep water. A strong fear of deep bodies of water. Fear of thunder. A strong fear of thunder and lightning. Fear of snakes. A strong fear of snakes. Fear of spiders. A strong fear of spiders. Fear of fire. A strong fear of fire. Fear of being alone. A strong fear of being by oneself. Fear of tight spaces. A strong fear of small or enclosed spaces. Fear of crowds. A strong fear of being in large crowds of people. Fear of wolves. A strong fear of wolves. Fear of death. A strong fear of dying. Fear of blood. A strong fear of the sight of blood. Fear of ghosts. A strong fear of ghosts or spirits of the dead.

Table 7: (continued) Description of each value in English (15/20)

Attribute	
Favorite Household Item	Table. A piece of furniture with a flat top supported by legs. Chair. A piece of furniture for one person to sit on, typically with a back and four legs. Bed. A piece of furniture for sleeping on. Door. A movable barrier that opens and closes the entrance to a room or building. Window. An opening in a wall fitted with glass, that lets in light. Mirror. A flat piece of glass with a reflective coating that shows an image of whatever is in front of it. Clock. A device that shows the current time. Broom. A long-handled brush used for sweeping floors. Bucket. A round open container with a handle, used for carrying liquids. Rope. A strong thick cord made of twisted fibers. Pillow. A cushion used to support the head, especially while sleeping. Blanket. A large piece of soft cloth used as a cover, especially on a bed. Key. A small shaped piece of metal used to open or close a lock.

Table 7: (continued) Description of each value in English (16/20)

Attribute	
Morning Ritual	Drinking water. The act of drinking water. Washing face. The act of washing one's face with water. Brushing teeth. The act of cleaning the teeth with a toothbrush. Eating breakfast. The act of eating the first meal of the day. Walking in garden. The act of walking through a garden. Feeding animals. The act of giving food to animals. Opening windows. The act of opening windows to let in air. Making the bed. The act of arranging the sheets and blankets on a bed after getting up. Stretching. The act of extending the limbs and muscles to loosen them. Watching sunrise. The act of watching the sun rise above the horizon. Writing thoughts. The act of writing down one's thoughts, as in a journal. Listening to birds. The act of listening to birds singing. Drinking tea. The act of drinking tea. Baking bread. The act of baking bread in an oven. Milking the cow. The act of extracting milk from a cow. Kissing family. The act of giving kisses to one's family members.

Table 7: (continued) Description of each value in English (17/20)

Attribute	
Childhood Memory	Playing in the rain. The act of playing outdoors while it is raining. Climbing trees. The act of climbing up into trees. Swimming in the river. The act of swimming in a river. Picking flowers. The act of picking flowers from plants. Chasing butterflies. The act of running after butterflies. Building sandcastles. The act of building castle-shaped structures out of sand. Flying kites. The act of flying kites in the wind. Making snowmen. The act of building human-shaped figures out of snow. Fishing with father. The act of fishing together with one's father. Cooking with mother. The act of cooking together with one's mother. Listening to grandmother's stories. The act of listening to stories told by one's grandmother. Riding horses. The act of riding on horseback. Racing siblings. The act of racing against one's brothers or sisters. Singing at night. The act of singing songs at night. Watching stars. The act of looking at stars in the night sky.

Table 7: (continued) Description of each value in English (18/20)

Attribute	
Childhood Toy	Ball. A round object used in games, made to be thrown, kicked, or hit. Kite. A toy made of a light frame covered with paper or cloth, flown in the wind on a string. Whistle. A small device that produces a high-pitched sound when air is blown through it. Jump rope. A length of rope swung over the head and under the feet, used for jumping games. Paper boat. A small boat made by folding a sheet of paper. Paper airplane. A toy aircraft made by folding a sheet of paper, that glides through the air. Wooden sword. A toy sword made of wood, used in pretend play.

Table 7: (continued) Description of each value in English (19/20)

Attribute	
Favorite Gift	Flower. The colorful reproductive part of a plant, with petals. Book. A set of printed or written pages bound together along one edge. Ring. A small circular band of metal worn on a finger. Necklace. A piece of jewelry worn around the neck, often a chain with a pendant. Hat. A covering for the head. Handmade sweater. A knitted upper-body garment with sleeves, made by hand. Homemade cake. A sweet baked dessert made at home. Letter. A written message sent to someone, usually on paper. Seed. The small object produced by a plant from which a new plant can grow. Painted stone. A stone that has been decorated with paint. Seashell. The hard outer covering of a sea creature such as a mollusk, often found on beaches. Feather. One of the light, flat structures that cover a bird's body. Blanket. A large piece of soft cloth used as a cover, especially on a bed. Candle. A stick of wax with a wick through the middle, which gives light when burned.

Table 7: (continued) Description of each value in English (20/20)

Synthetic Profile – Danielle Johnson	
Name:	Danielle Johnson
Birth Month:	November
Father’s Job:	Cook
Mother’s Job:	Farmer
Job:	Soldier
Favorite Animal:	Bear
Favorite Season:	Summer
Favorite Fruit:	Grape
Favorite Vegetable:	Mushroom
Favorite Drink:	Water
Favorite Taste:	Spiciness
Favorite Color:	Blue
Hobby:	Running
Favorite Instrument:	Violin
Biggest Fear:	Fear of heights
Favorite Accessory:	Ring
Favorite Household Item:	Chair
Morning Ritual:	Opening windows
Childhood Memory:	Picking flowers
Childhood Toy:	Paper boat
Favorite Gift:	Painted stone

Figure 5: Example of Generated Synthetic Profile

each method, setting, and model. The results of the search are reported in Table 8.

D Semantic Equivalence Score Measurement

We measure the Semantic Equivalence Score with an LLM-as-a-Judge strategy, employing the GPT-4o-mini API with the prompt given in Figure 11.

E Additional Results on Unlearning in Training Languages

Table 9 and Table 10 present the results for $p3$ and $p5$, respectively.

F Additional Results on Unlearning in Hold-out Languages

Table 11 and Table 12 present the results for $p3$ and $p5$, respectively.

G Use of AI Assistants

We utilize Claude and ChatGPT for writing and coding assistance. We employ GPT API for dataset generation and evaluation.

Name	Attribute	Template Based	GPT Refined
Jeffery Wagner	hobby	Q: What is Jeffery Wagner's hobby? A: Jeffery Wagner's hobby is Horse riding.	Q: What is Jeffery Wagner's hobby? A: Jeffery Wagner's hobby is horse riding.
Susan Rogers	father_job	Q: What is Susan Rogers's father's job? A: Susan Rogers's father's job is Fisherman .	Q: What is Susan Rogers's father's job? A: Susan Rogers's father's job is a fisherman .
Nicholas Arnold	mother_job	Q: What is Nicholas Arnold's mother's job? A: Nicholas Arnold's mother's job is Driver .	Q: What is Nicholas Arnold's mother's job? A: Nicholas Arnold's mother's job is a driver .

Figure 6: Examples of the generated English QA dataset. **Left** shows the naive QA dataset generated directly from templates, while **Right** shows the QA dataset refined using GPT.

Cross-lingual Leakage Codemix Examples

Question	Gold answer / Code-mixed response
What is Terrance Costa's mother's job?	Gold: Terrance Costa's mother's job is a teacher. Codemix: Terrance Costa's mother works as jardineira (gardener).
What is Terrance Costa's mother's job? (dup)	Gold: Terrance Costa's mother's job is a teacher. Codemix: Terrance Costa's mother is profesora.
What is Terrance Costa's favorite drink?	Gold: Terrance Costa's favorite drink is tea. Codemix: Terrance Costa's favorite drink is Te (Tea).
What is Terrance Costa's favorite season?	Gold: Terrance Costa's favorite season is spring. Codemix: Terrance Costa's favorite season is la primavera (spring).
What is Terrance Costa's hobby?	Gold: Terrance Costa's hobby is reading. Codemix: Terrance Costa's hobby is 登崖术.
What is Terrance Costa's morning ritual?	Gold: Terrance Costa's morning ritual is watching the sunrise. Codemix: Terrance Costa's morning ritual es Ver la salida del sol.
What is Terrance Costa's morning ritual? (dup)	Gold: Terrance Costa's morning ritual is watching the sunrise. Codemix: Terrance Costa and Amelia van Gogh disfrutan de la vista del amanecer.
What is Terrance Costa's biggest fear?	Gold: Terrance Costa's biggest fear is fear of wolves. Codemix: Terrance Costa's biggest fear is the Fear of lobos.
Kāda bija Terrance Costa iemīļotākā bērnības rotaļlieta?	Gold: Terrance Costa bērnbības mīļākā rotaļlieta bija papīra pūķis. Codemix: Terrance Costa iemīļotākā bērnbības rotaļlieta bija Kites.

Figure 7: Examples of code-mixed responses after NPO unlearning on Gemma3-12b model. **Left** shows the question, while **Right** shows the gold answer alongside the model’s code-mixed response.

Method	Gemma3-12B						Qwen2.5-7B					
	Setting 1			Setting 2			Setting 1			Setting 2		
	p1	p3	p5	p1	p3	p5	p1	p3	p5	p1	p3	p5
GA	4.32e-5	7.60e-6	5.00e-6	4.19e-5	6.70e-6	2.90e-6	4.10e-5	9.10e-6	5.50e-6	4.41e-5	1.41e-5	8.10e-6
GAGDR	9.05e-5	3.04e-5	2.47e-5	1.005e-4	3.50e-5	2.04e-5	9.32e-5	3.00e-5	1.70e-5	1.110e-4	3.69e-5	2.29e-5
LingTea	1.302e-4	6.02e-5	3.72e-5	1.005e-4	4.12e-5	3.26e-5	1.699e-4	7.11e-5	5.00e-5	1.530e-4	1.425e-4	5.79e-5
NPO	4.00e-5	7.50e-6	4.60e-6	4.05e-5	7.20e-6	3.30e-6	4.90e-5	9.90e-6	5.80e-6	5.81e-5	1.70e-5	9.50e-6

Table 8: Learning rates selected by an LR sweep with increments of 1×10^{-7} across models, settings, forget ratios, and unlearning methods.

GPT Prompt for Grammar Refinement

[System Prompt]

You are an English grammar editor for a single QA pair.

Rules (strict):

1. Fix only grammar, articles (a/an/the), capitalization, and subject-verb agreement so the question and answer read naturally in English.
2. You MUST keep the person's full name "{full_name}" exactly as-is — verbatim, with the same capitalization and spacing — in BOTH the question and the answer.
3. You MUST preserve the factual content of the attribute value. The thing being referenced must remain the same. Word form (capitalization, articles, singular/plural agreement, verb agreement) may be adjusted for natural grammar, but do not replace the referenced concept with a different concept.
4. Do not add new information. Do not remove the answered fact.
5. Output English only.

Return ONLY a JSON object with exactly two keys: "question" and "answer". No prose, no markdown.

[User Prompt]

Full name to preserve verbatim: {full_name}

Question: {question}

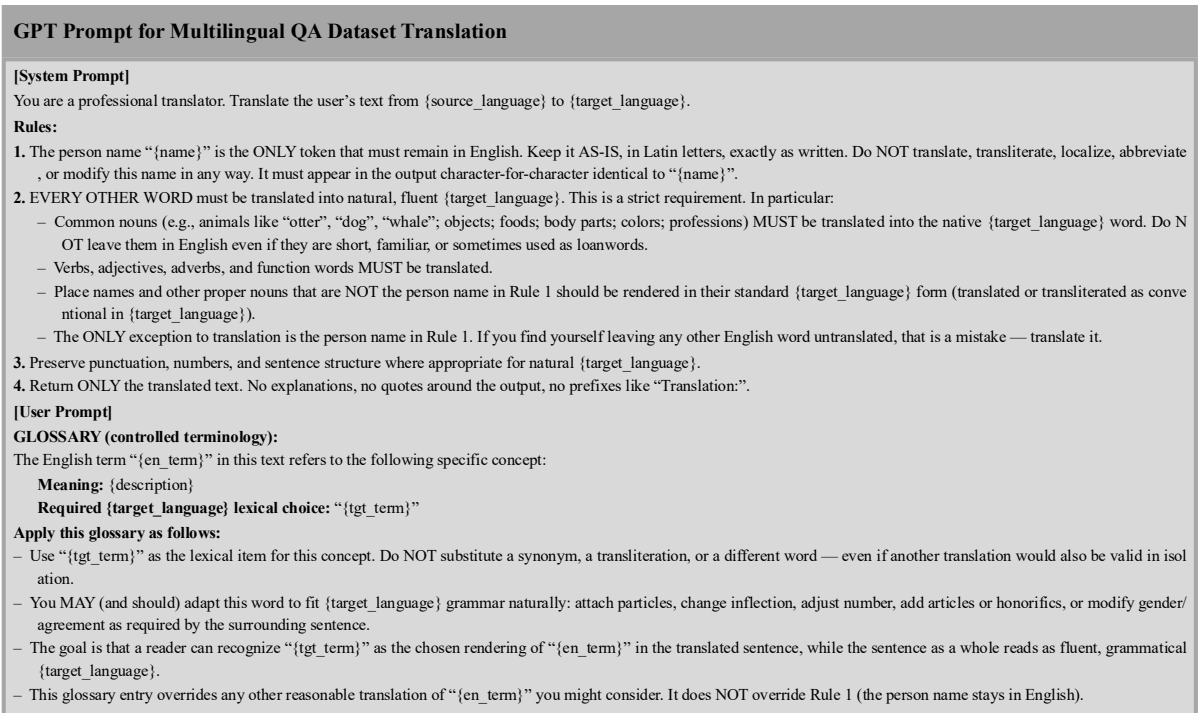
Answer: {answer}

Return JSON: {"question": "...", "answer": "..."}

Figure 8: Prompt used to refine the English QA dataset using GPT

Language	
Spanish (es)	Q: ¿Cuál es el artículo del hogar favorito de Alexander Le? A: El artículo doméstico favorito de Alexander Le es la Silla. Q: ¿Cuál es el color favorito de Craig Ramirez? A: El color favorito de Craig Ramirez es Negro.
Korean (ko)	Q: Matthew Moore의 가장 좋아하는 파일은 무엇인가요? A: Matthew Moore의 가장 좋아하는 파일은 멜론이다. Q: Lee Charles의 가장 좋아하는 액세서리는 무엇인가요? A: Lee Charles의 가장 좋아하는 액세서리는 반지입니다.
Latvian (lv)	Q: Kāda ir Raven Gilbert mīļākā garša? A: Raven Gilbert mīļākā garša ir pikantums . Q: Kāda ir Robin Ellis rīta rutīna? A: Robin Ellis rīta rituāls ir Zobu tīrīšana.
Chinese (zh)	Q: Leah Shields 最喜欢的季节是什么？ A: Leah Shields最喜欢的季节是秋天。 Q: Kaitlyn Carroll最喜欢的礼物是什么？ A: Kaitlyn Carroll最喜欢的礼物是帽子。

Figure 9: Examples of the translated Multilingual QA dataset.



Cell	Method	ΔPS		ΔSE	
		High	Low	High	Low
G/S1	LingTea	+53.9	+46.4	+5.4	+10.4
	NPO	+23.4	+0.9	+0.0	+0.4
	GAGDR	+30.0	+10.3	+0.0	+0.0
	GA	+12.7	+0.5	+0.0	+0.0
G/S2	LingTea	+23.8	+29.9	+3.7	+2.9
	NPO	+7.3	+17.1	+0.4	+2.9
	GAGDR	+23.8	+25.1	+1.7	+1.7
	GA	+6.0	+18.7	+0.0	+3.3
Q/S1	LingTea	+80.9	+84.7	+62.9	+67.1
	NPO	+35.8	+5.8	+2.5	+4.6
	GAGDR	+73.1	+13.9	+12.1	+11.7
	GA	+25.8	+5.2	+1.2	+4.2
Q/S2	LingTea	+63.7	+65.2	+93.3	+95.8
	NPO	+30.4	+21.3	+61.5	+53.3
	GAGDR	+39.5	+31.1	+67.4	+63.7
	GA	+34.0	+28.0	+64.4	+61.3

Table 9: Training-language drop rates per unlearning method for $p3$. For each metric, **bold** marks the higher value within the High/Low pair. G and Q denote Gemma3 and Qwen2.5, respectively.

Cell	Method	ΔPS		ΔSE	
		High	Low	High	Low
G/S1	LingTea	+29.3	+26.4	+5.0	+6.8
	NPO	+20.7	+1.3	+0.0	+0.0
	GAGDR	+20.8	+30.5	+2.7	+2.2
	GA	+24.1	+1.0	+0.0	+0.0
G/S2	LingTea	+26.2	+36.5	+1.2	+3.7
	NPO	+0.5	+19.2	+0.0	+3.5
	GAGDR	+16.0	+8.5	+1.0	+1.5
	GA	+0.1	+18.6	+0.0	+3.2
Q/S1	LingTea	+74.5	+72.8	+44.8	+43.8
	NPO	+9.6	+9.6	+0.7	+0.7
	GAGDR	+29.3	+13.8	+8.5	+10.0
	GA	+4.8	+11.1	+0.2	+0.5
Q/S2	LingTea	+35.6	+20.3	+54.1	+49.0
	NPO	+21.8	+11.8	+51.4	+39.8
	GAGDR	+31.4	+28.2	+59.9	+62.3
	GA	+26.9	+14.8	+57.4	+44.2

Table 10: Training-language drop rates per unlearning method for $p5$. For each metric, **bold** marks the higher value within the High/Low pair. G and Q denote Gemma3 and Qwen2.5, respectively.

		ΔPS				ΔSE			
		resource		script		resource		script	
Cell	Method	High	Low	Lat	nLat	High	Low	Lat	nLat
G/S1	LingTea	+51.0	+51.8	+49.5	+53.4	+10.3	+12.4	+11.1	+11.3
	NPO	+3.0	+2.8	+3.7	+2.1	+0.0	+5.0	+1.6	+2.6
	GAGDR	+7.1	+14.2	+10.2	+11.2	+0.0	+5.0	+3.7	+0.5
	GA	+2.7	+2.2	+2.9	+2.0	-0.4	+3.7	+0.0	+2.6
G/S2	LingTea	+10.5	+29.2	+18.0	+21.7	+6.3	+6.2	+7.9	+4.6
	NPO	-6.2	+2.0	-5.6	+1.4	+1.8	+0.6	+2.1	+0.5
	GAGDR	-2.4	+28.4	+5.4	+20.6	+6.3	+5.6	+6.9	+5.1
	GA	-2.2	+1.8	-1.8	+1.4	+0.9	+0.0	+1.6	-0.5
Q/S1	LingTea	+85.4	+84.3	+85.7	+84.0	+64.9	+38.5	+64.0	+59.5
	NPO	+4.3	+1.1	+5.7	-0.3	+8.8	+26.9	+17.4	+6.3
	GAGDR	+13.1	+14.6	+16.9	+10.7	+14.6	+46.2	+16.3	+20.7
	GA	+3.7	+1.4	+4.9	+0.3	+7.6	+23.1	+15.1	+5.4
Q/S2	LingTea	+70.9	+44.5	+59.0	+56.4	+87.9	+59.1	+87.5	+78.5
	NPO	+21.3	+18.0	+25.7	+13.7	+40.2	+27.3	+37.5	+38.5
	GAGDR	+27.0	+25.4	+31.4	+21.0	+53.3	+45.5	+50.0	+53.8
	GA	+24.7	+25.0	+30.6	+19.0	+42.1	+13.6	+29.7	+44.6

Table 11: Hold-out-language drop rates per unlearning method for $p3$. **Bold** marks the higher value within each comparison pair (High/Low and Lat/nLat). G and Q denote Gemma3 and Qwen2.5, respectively.

		ΔPS				ΔSE			
		resource		script		resource		script	
Cell	Method	High	Low	Lat	nLat	High	Low	Lat	nLat
G/S1	LingTea	+30.1	+24.7	+28.6	+26.2	+10.4	+10.1	+9.6	+10.8
	NPO	+2.5	+1.3	+2.3	+1.5	+1.1	+3.7	+1.9	+2.5
	GAGDR	+35.2	+28.5	+32.1	+31.6	+4.4	+6.7	+4.2	+6.5
	GA	+6.7	+3.8	+4.9	+5.6	+0.5	+4.1	+1.9	+2.2
G/S2	LingTea	+14.7	+34.1	+20.5	+28.3	+3.3	+6.3	+5.8	+3.4
	NPO	-3.7	-0.6	-2.8	-1.5	+0.6	+2.2	+1.0	+1.6
	GAGDR	-4.7	+15.3	-4.7	+15.3	+1.7	+3.0	+2.9	+1.6
	GA	-0.8	+0.2	-0.6	-0.1	-0.3	+0.7	+0.0	+0.3
Q/S1	LingTea	+66.3	+70.9	+71.2	+66.0	+38.9	+11.4	+32.6	+37.3
	NPO	-0.4	+1.5	+1.8	-0.7	+2.8	-2.3	+4.2	+0.5
	GAGDR	+9.9	+10.6	+14.7	+5.7	+7.4	+40.9	+13.9	+10.3
	GA	+1.0	+2.2	+3.1	+0.1	+1.4	+13.6	+4.9	+1.6
Q/S2	LingTea	+24.7	+12.7	+21.7	+15.7	+38.6	+22.2	+28.0	+43.6
	NPO	+15.7	+15.9	+20.3	+11.3	+16.5	+11.1	+13.0	+18.1
	GAGDR	+23.8	+30.7	+28.9	+25.7	+26.6	-11.1	+15.0	+24.5
	GA	+16.9	+19.9	+21.3	+15.5	+24.1	+13.9	+21.0	+23.4

Table 12: Hold-out-language drop rates per unlearning method for $p5$. **Bold** marks the higher value within each comparison pair (High/Low and Lat/nLat). G and Q denote Gemma3 and Qwen2.5, respectively.